WORKSHEET

Expected values using a calculator

1 Use a calculator to determine the expected value of the following distributions.

a

x	1	2	3	4
$P(X = x)$	0.1	0.3	0.2	0.4

b

y	1	2	3	4
$P(Y = y)$	0.25	0.25	0.25	0.25

c

z	35	38	40	41	44	46
$P(Z = z)$	0.12	0.05	0.34	0.15	0.21	0.13

d

w	-2	-1	0	1	2
$p(w)$	0.354	0.111	0.243	0.099	0.193

e

x	0	25	80	135	190	265
$p(x)$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{24}$	$\frac{1}{12}$	$\frac{3}{8}$	$\frac{1}{6}$

f

x	1	2	3	4	5	6	7
$p(x)$	0.5	0.25	0.125	0.0625	0.03125	0.015625	0.015625

g

y	0	2	4	6	8
$P(Y = y)$	0.4	0.2	0.2	0.05	0.15

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h

x	-12	-7	-3	2	5	9
$P(X = x)$	0.281	0.18	0.119	0.23	0.08	0.11

2 A coin is tossed three times and random variable X = the number of heads recorded.

a Complete the table.

x				
$P(x)$				

b Use a calculator to determine $E(X)$.

3 A coin is tossed three times and random variable Y = the number of tails in a row.

a Complete the table.

y				
$P(y)$				

b Use a calculator to determine $E(Y)$.

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Answers

1 a 2.9

b 2.5

c 41.07

d -0.334

e 136.25

f ≈ 1.984

g 2.7

h -3.139

2 a

x	0	1	2	3
$P(x)$	0.125	0.375	0.375	0.125

b 1.5

3 a

y	0	1	2	3
$P(y)$	0.125	0.5	0.25	0.125

b 1.375

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